

Miaohui Shi / 施妙辉

Algorithm Engineer | High-performance Computing & Robotics

Taizhou, Zhejiang, China | <https://shimiaohui0426.github.io/cv/>

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Algorithm engineer at Taizhou Optoelectronics Industry Innovation Center. M.S. in Mechanical Engineering, Waseda University, completed March 2024. Research and engineering in GDS rasterization, multi-robot interaction and vision-based tactile sensing.

PROFESSIONAL EXPERIENCE

Algorithm Engineer | Sep 2024 - present

Taizhou Optoelectronics Industry Innovation Center / 台州光电产业创新中心

- Develop EDA-related algorithms and software using C++ and Python. Ported GDS rasterization from Python to C++ and optimized computation, parallelism, memory access and compilation.
- Migrated the workload from a small PC to multi-node servers. Full progression: an initial estimate of tens of thousands of hours to a measured 24-minute six-node run. Measured 800 to 24 minutes gives 33.3× end-to-end acceleration, including hardware and deployment changes.
- Implemented partitioned reads so each node loads only its assigned data. On the same server configuration and node count, reduced processing from 92 to 44 minutes (2.09×). Minute-level comparisons use the same GDS input and 4-pass workload.
- Extended the pipeline with multi-node processing, CD bias and correction support; built software for local inspection of very large raster images and equipment control.

Algorithm Development | Apr - Aug 2024

Hesai Technology / Sharp-related R&D

- Formal employer: Hesai Technology. Worked on LiDAR functional-degradation algorithms and noise-segmentation model training, alongside Sharp-related tactile-sensor R&D.
- Designed and trained mechanical models for vision-based tactile sensors; developed and trained models on Ubuntu.

EDUCATION

M.S., Mechanical Engineering | 2022–Mar 2024

Waseda University · Japan

B.S., Mechanical Science & Engineering | 2017–2021

Huazhong University of Science and Technology · China

TECHNICAL SKILLS

Performance & engineering: C++ · Python · Linux / Ubuntu · parallel and distributed processing · memory and I/O optimization · GDS rasterization

Vision & learning: YOLO · semantic segmentation · anomaly detection · model training and optimization · vision-based tactile sensing

Robotics & interaction: ROS · Unity · multi-robot systems · HRI experiment design · PASAT / NASA-TLX · behavioral data analysis

Software & interfaces: C# · Qt · front-end UI development · industrial vision software · TwinCAT / PLC integration

EARLIER ACADEMIC EXPERIENCE

Oct–Dec 2022 | Learning Assistant · Waseda University

Dec 2019–Feb 2022 | Research Assistant · Huazhong University of Science and Technology

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SELECTED RESEARCH

Multi-robot interaction, attention and automation | CASE 2024

Independently developed the experimental system (excluding the mechanical-head module), designed and conducted the experiments, and completed all data analysis. Adapted the PASAT auditory-addition task to a dual-task study involving three robotic arms, with NASA-TLX and user-experience measures.

High-speed vision-based tactile sensing | ICAT-EGVE 2024

Independently developed algorithmic principles, implementation, performance tuning and the front-end UI. The method combines local sparse marker tracking with force estimation using an asymmetric stiffness-coefficient matrix. The paper reports 601.25 Hz sensor-system force acquisition; this is not a standalone algorithm benchmark.

PUBLICATIONS

2024 | High-Speed Vision-Based Haptic Sensor for Robotic Dermatological Palpation: Force Sensing Method Using Asymmetric Stiffness Coefficient Matrix

Fumihiro Kato, Miaohui Shi, Kaito Kamishima, Hiroyasu Iwata

ICAT-EGVE 2024 | DOI: [10.2312/egve.20241368](https://doi.org/10.2312/egve.20241368)

2024 | The Role of automation and User Experience in Multi Robot Systems

Kouta Suzuki, Miaohui Shi, Vimolmongkolporn Vitvasin, Yukiko Iwasaki, Hiroyasu Iwata

IEEE CASE 2024 · pp. 3818–3825 | DOI: [10.1109/CASE59546.2024.10711808](https://doi.org/10.1109/CASE59546.2024.10711808)

2023 | Balancing Real-world Interaction and VR Immersion with AI Vision Robotic Arm

Xiaoran Yang, Yang Zhan, Yukiko Iwasaki, Miaohui Shi, Shijie Tang, Hiroyasu Iwata

IEEE ICMA 2023 · pp. 2051–2057 | DOI: [10.1109/ICMA57826.2023.10215826](https://doi.org/10.1109/ICMA57826.2023.10215826)

2021 | 基于TwinCAT的机器人实验平台设计

黄骏, 施妙辉, 汪迪, 王峻峰

实验技术与管理 · 38(12):197–202+207 | DOI: [10.16791/j.cnki.sjg.2021.12.039](https://doi.org/10.16791/j.cnki.sjg.2021.12.039)

2021 | A Remote Experiment Platform based on HTML5 for Industrial Robots

Tao Huang, Miaohui Shi

MED 2021 · pp. 920–923 | DOI: [10.1109/MED51440.2021.9480211](https://doi.org/10.1109/MED51440.2021.9480211)

PATENTS

Eight distinct patent application records are listed on the website, covering visual inspection, material handling, subpixel image processing and vision-based tactile sensing. Publication and grant versions of the same application are counted once.

<https://shimiaohui0426.github.io/patents/>

RESEARCH DIRECTION

Future doctoral research interest: household service robots, tactile perception, human-robot collaboration and appropriate automation. Industry interests include EDA, computational lithography and high-performance algorithm engineering.